Kursintroduktion EL1000 Reglerteknik AK P1 2026

Kursansvarig: Angela Fontan
Föreläsare F1-F2: Henrik Sandberg

Reglerteknik (DCS), EECS
KTH Royal Institute of Technology, Sweden

Reglerteknik EL1000, P1 2026

Se till att du är kursregistrerad!

- Webpage: https://canvas.kth.se/courses/35047, see also Kursinformation 2026
- Administration: Contact EECS Studentexpeditionen
- Programs: Teknisk Fysik + Teknisk Matematik + Energi och Miljö + ...
- Kursen ger totalt 6 hp:
  - Lab 1: 0.5hp
  - Lab 2: 1.5hp
  - Lab 3: 2hp
  - Tentamen: 2hp
- Vi söker kursnämndsrepresentanter för Teknisk fysik och Teknisk matematik – kontakta Angela (angfon@kth.se)

Lectures and Course Literature

- Lecturers:
  - Henrik Sandberg, hsan@kth.se, F1–F2
  - Angela Fontan, angfon@kth.se, F3–F12

  Lectures are given in English on the whiteboard. Slides are uploaded on Canvas before the lectures (Kursmaterial finns som pdf-filer under modulen Kursmaterial)

- Course Literature:
  Lennart Ljung, Torkel Glad och Martin Enqvist: *Reglerteknik, Grundläggande teori,* (Upplaga: Femte)

  English/Swedish alternative:
  A. Hansson, Control Engineering: An Introduction, BoD, 2025.
  - Good complementary material: Franklin, Powell and Emami-Naeini, "Feedback Control of Dynamic Systems", Sixth Edition, Pearson, 2010; Åström and Murray, "Feedback Systems: An Introduction for Scientists and Engineers"
  - Other material: engelska svenska reglerteknik-ordlistan in Kursmaterial

Exercise sessions and TAs

- Dennis Bogatov Wilkman, Grupp A på svenska, dwilkman@kth.se
- Yulun Wu, Group B in English, yulunw@kth.se
- Sara Betancur Giraldo, Group C in English, sarabg@kth.se
- Jacopo Porzio, Group D in English, porzio@kth.se

Räknestugor/Math help sessions: All

Laboratories: LAB 1 and LAB 2

4h laboratory work in **groups of 2 people**

- LAB 1 starts next week on Monday, August 31!
- Where: Reglertekniks undervisningslab at Malvinas väg 10, floor 2
- LAB 1: Preparation via lab exercises. **Mandatory Quiz after the lab.**
- LAB 2: **Mandatory Quiz before the lab**, based on 5 tests available in Kursmaterial
- Registration for the labs (LAB 1 and LAB 2) is done via Canvas/People/Groups. To register, you must be enrolled in the course.

Registration opens on August 26

- Registration for LAB 1 must be made no later than Sunday, August 30
- Registration for LAB2 must be made no later than Sunday, September 20

After these dates, no places can be guaranteed

Information + mandatory quizzes can be found under "Laborationer och Datorprojekt"

Laboratories: LAB 3

LAB 3: A computer project in **groups of 2 people**

- Registration for the presentation (LAB 3) is done via Canvas/People/Groups. To register, you must be enrolled in the course.
- Study a control problem in detail in MATLAB
- NOTE! Completed “dator” exercises (Ö4, Ö7, Ö11) will help
- Start working on LAB 3 well in advance! It is advisable to start after Lecture 5
- Presentation (15 min/group) in Zoom with shared screen

Information can be found under “*Laborationer och Datorprojekt / Laboration 3 (OBS kräver föranmälan och förberedelse)*”

The exam

- På svenska
- Torsdagen den 15 oktober 2026 kl.8.00–13.00
- **Anmälan senast två veckor innan via “Mina Sidor”**
- Kursbok är tillåten. Övningar, labs och extentor mm är ej tillåtna.

Course contents

- How feedback influences properties of dynamic systems, such as stability, speed of response, sensitivity, and robustness
- Analysis and design of feedback systems with regard to these properties

In particular:

- *Basic concepts and problems:* representation of dynamic systems, input and output signals, differential equation models, transforms, transfer functions, block diagrams, impulse response, step response, poles, zeros, and linearization
- *Analysis of feedback systems:* stability, root locus, the Nyquist criterion, Nyquist and Bode diagrams, precision, speed of response, sensitivity, and robustness
- *Design of SISO control systems:* specifications, PID-controllers, compensation in the frequency domain, feed-forward control, time delays, state feedback, observers, and pole placement
- *Implementation:* choice of sampling time, anti-alias filters, and discretization of controllers
- *Control terminology* in Swedish and English.